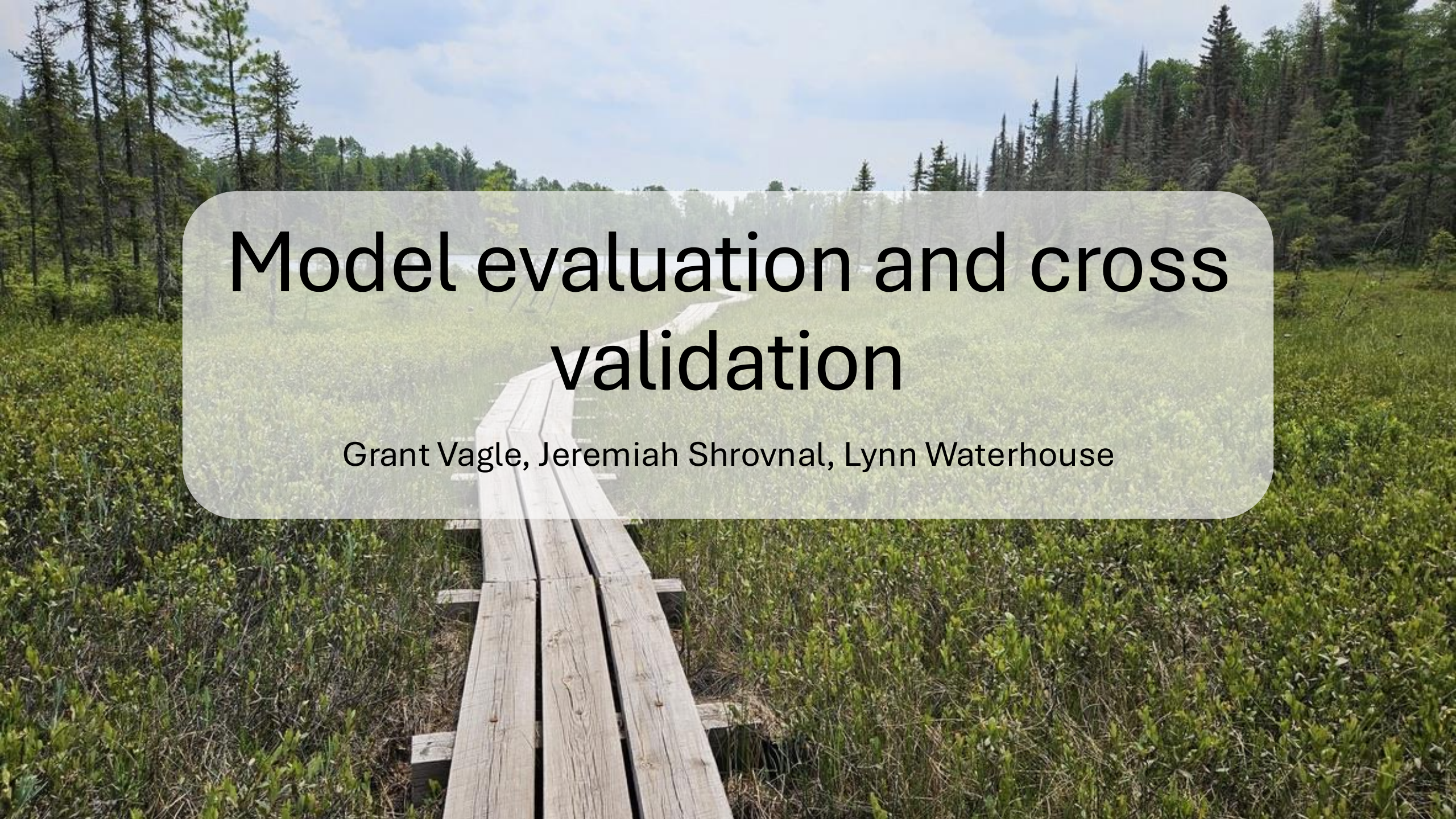


Model evaluation and cross validation

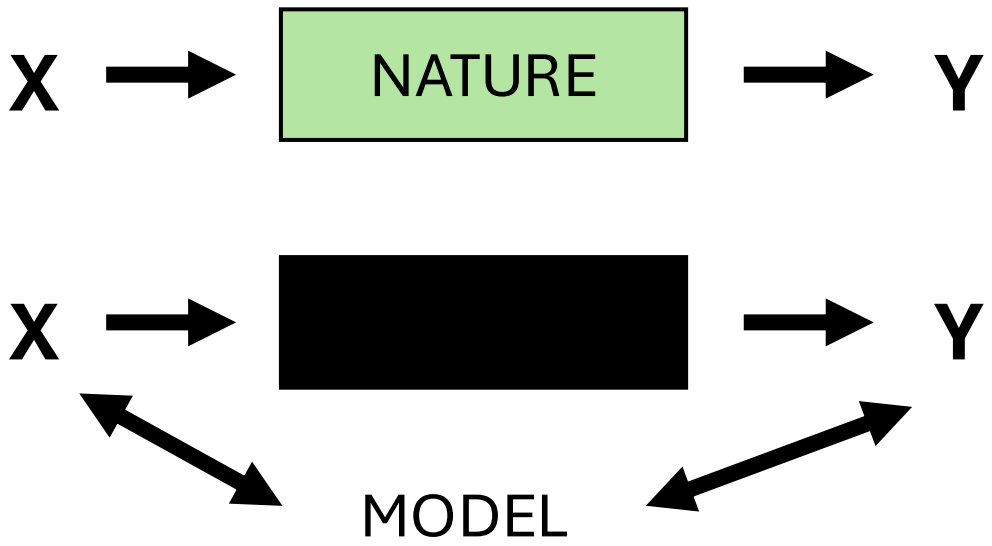
Grant Vagle, Jeremiah Shrovnal, Lynn Waterhouse

Overview of this section

- Understanding how well a model is performing
  - Performance metrics
  - Evaluation / cross-validation
- Introduce Classification and Regression Trees (CART)
- Hands-on coding example

How do we measure how well a model mimics the outputs of nature?

1. Model performance metrics – *what* we calculate
2. Model validation – *which data* we use to make the calculation



Molnar, C. 2023. Modeling mindsets: the many cultures of learning from data.

Model performance metrics - classification

Many (most?) performance metrics for classification come from the **confusion matrix**

Let's say we have a model that predicts a yes or no response

	Observed response	Predicted response
1	Yes	Yes
2	No	Yes
3	Yes	Yes
…		
100	Yes	No

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Metrics from confusion matrix:
https://www.geeksforgeeks.org/machine-learning/confusion-matrix-machine-learning/

Metrics for classification:
https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Link to ROC and AUC details:
https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

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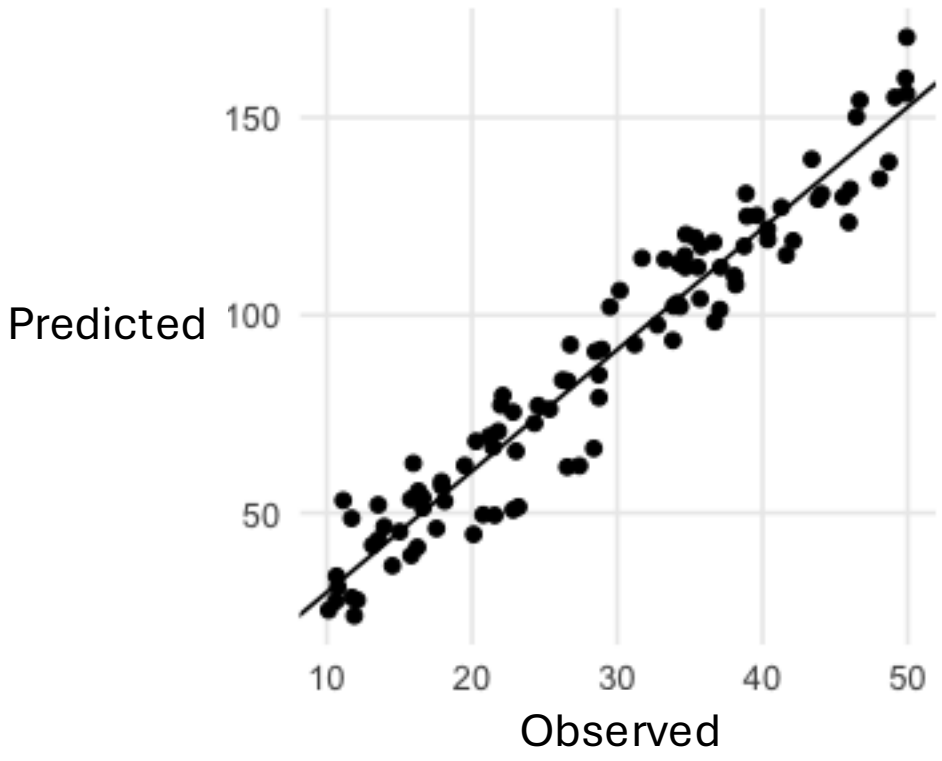
Model performance metrics - regression

Most ask in some way: **how big are the residuals?**



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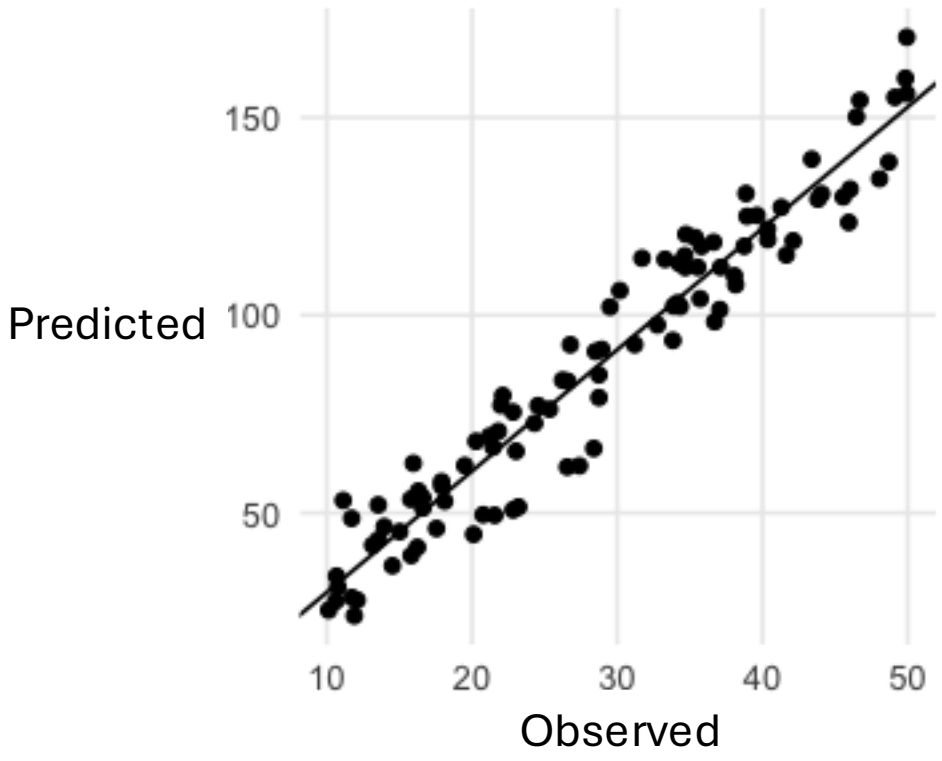

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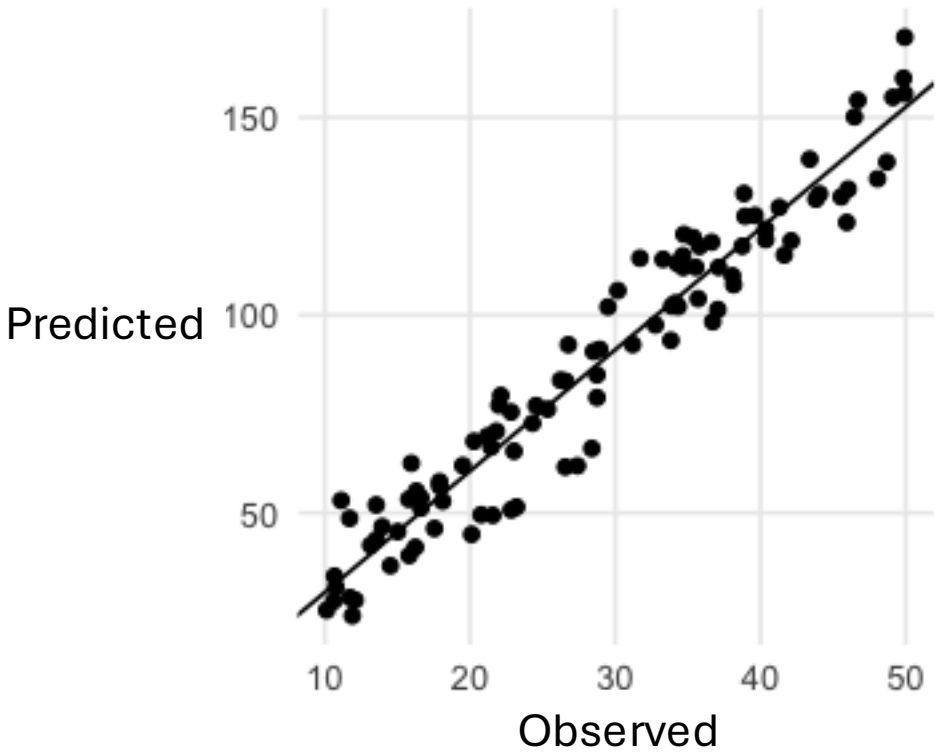

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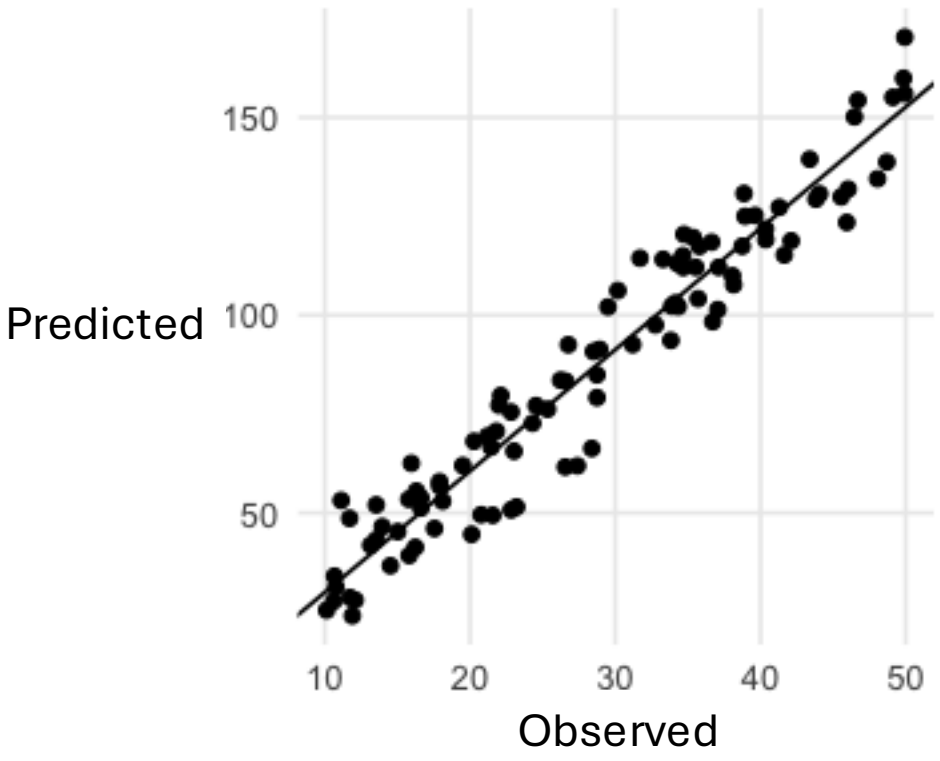

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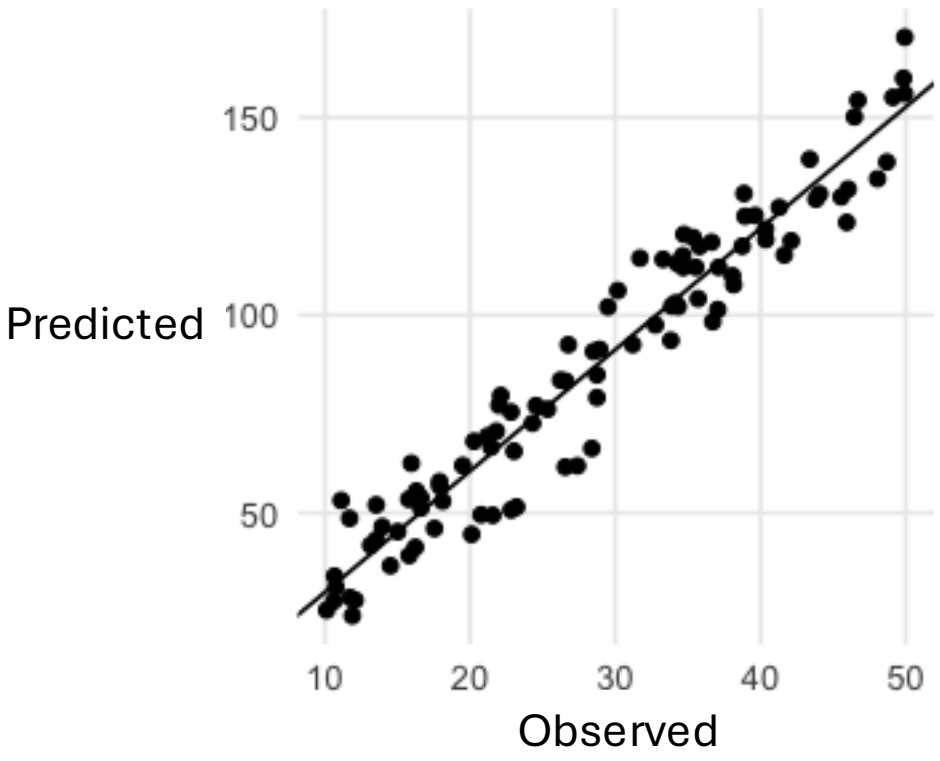


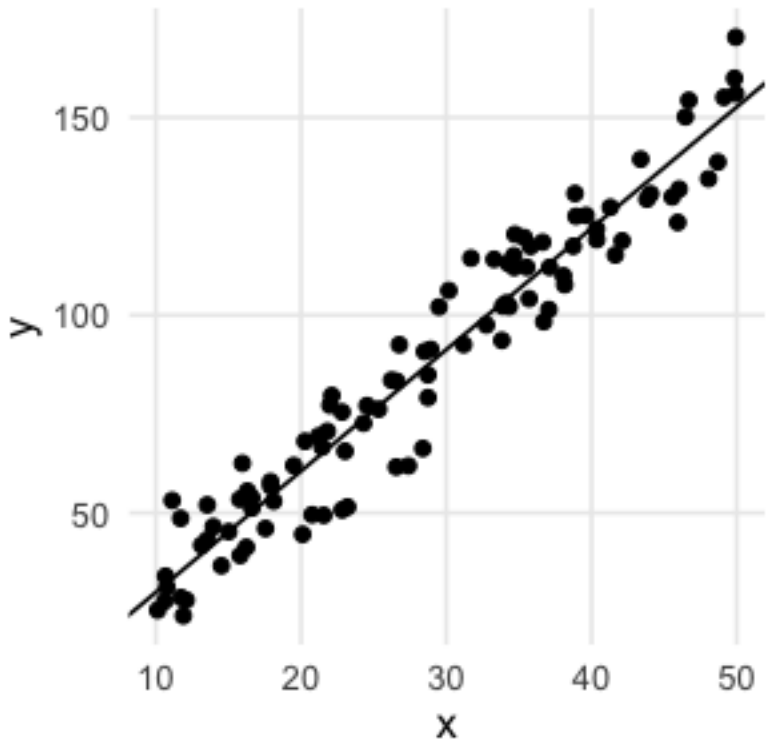
$$RMSE = \frac{\sqrt{\sum(observed - predicted)^2}}{\mathrm{n}}$$

$$\mathrm{R}^2 = 1 - \frac{\mathrm{RMSE}^2}{\mathrm{Var(y)}}$$

RMSE vs R^2
https://datascience.stackexchange.com/questions/100605/rmse-vs-r-squared

Model performance metrics - regression

- Many metrics compute some sort of “average” to get at overall performance



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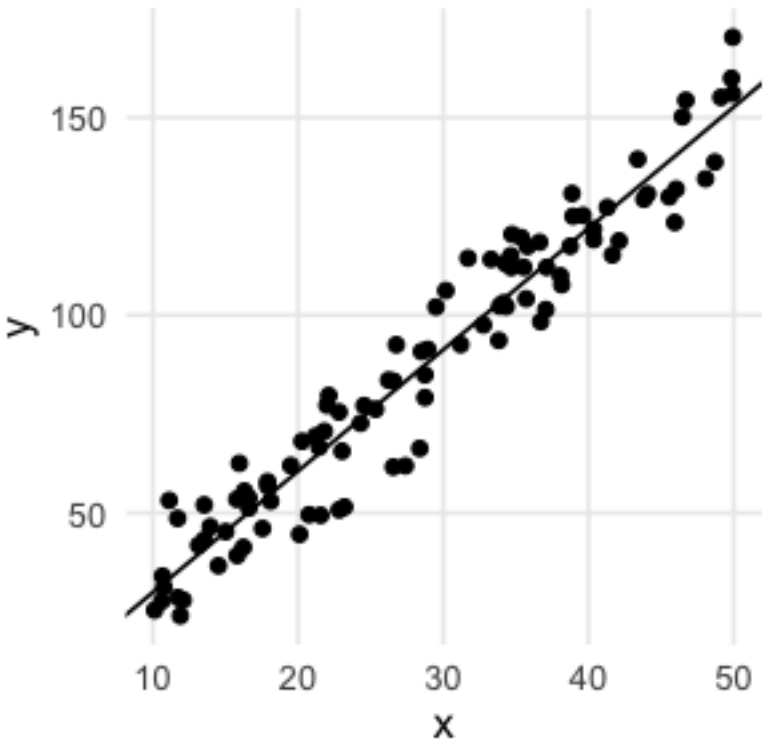

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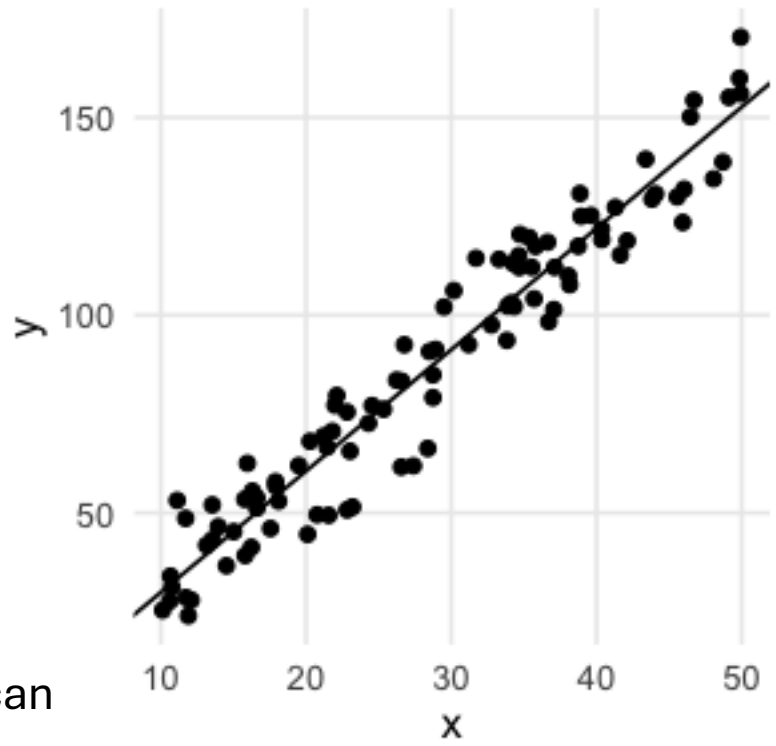

Here’s a link with lots of metrics that can be calculated, with some practical examples: https://towardsai.net/p/artificial-intelligence/the-essential-guide-to-ml-evaluation-metrics-for-regression

Model performance metrics – key takeaways

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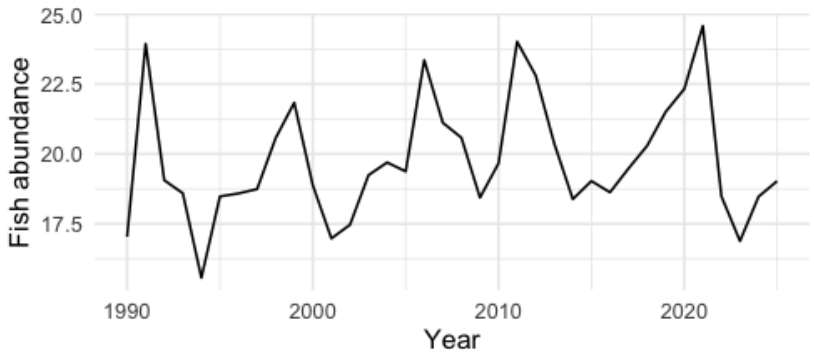
Because the model is completely focused on mimicking the outputs, how you measure how well it mimics them is highly important

An example…

We build a model to predict a lake's fish abundance over time based on fishing pressure and weather.

We train the model using data from 1990-2025



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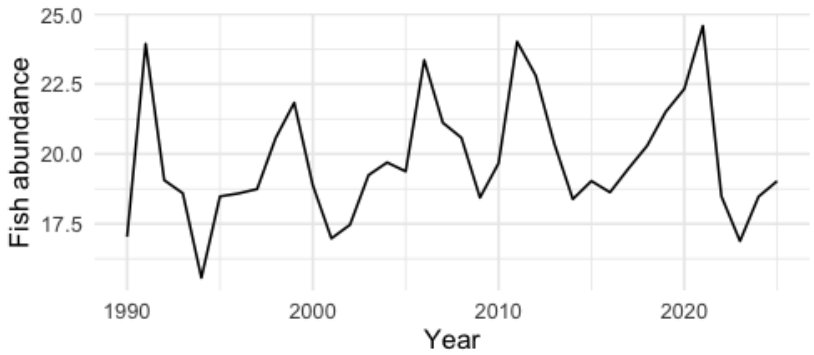

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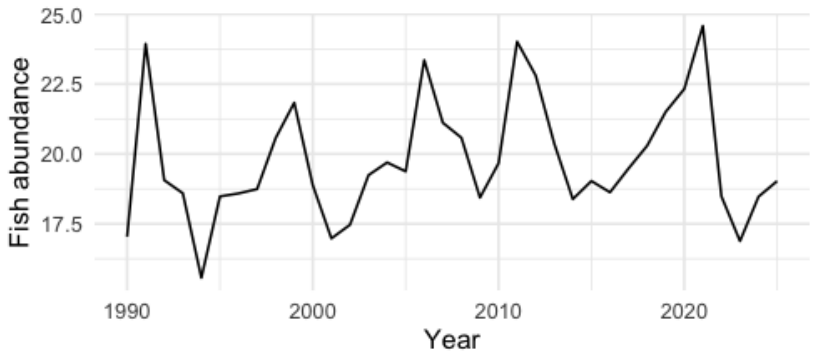

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3 sets of data

Training set: used to train the model

Validation set: used by the model/researcher to adjust parameters

Test set: used to evaluate the model performance

3 sets of data

Often handled internally in the code packages that build the models

Training set: used to train the model

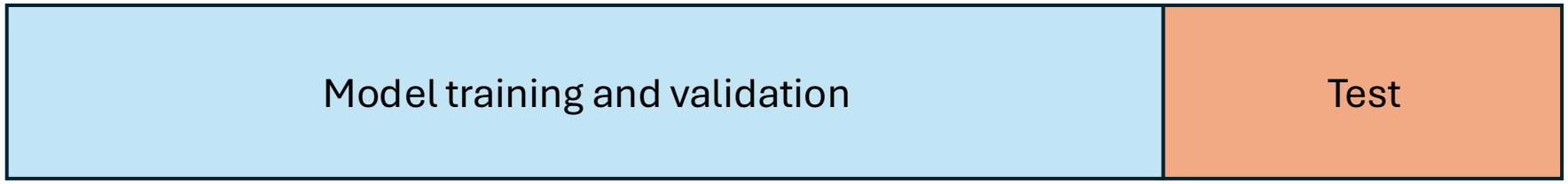
Validation set: used by the model/researcher to adjust parameters

Test set: used to evaluate the model performance

You should *always* evaluate your model with an **independent test set**

The simplest option is to:

- Hold out 20% of your dataset as the **test set**
- Train your model on the 80% **training set**
- Calculate performance metrics on the test set

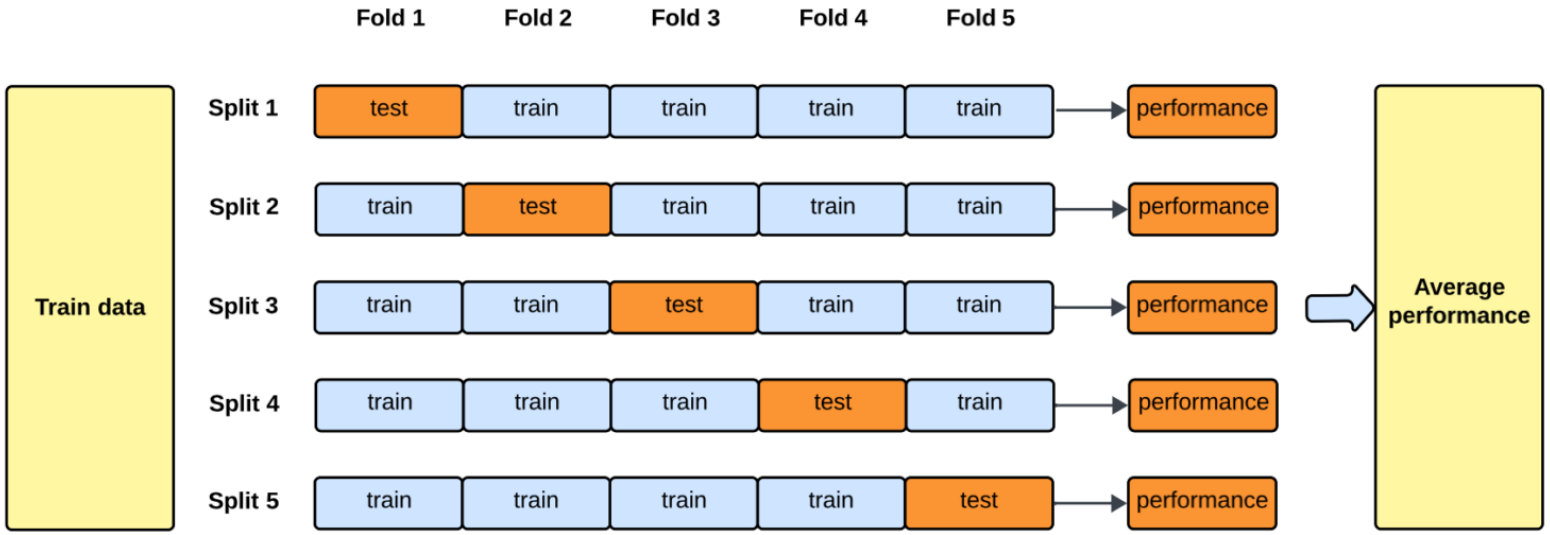


Sometimes, sample size makes this impractical or requires alternatives for a specific application…we'll talk about those later.

K-fold cross-validation

Randomly split your training data into k subsets (“folds”)

For k = 5:



When k = number of rows, it’s a leave-one-out cross validation

https://towardsdatascience.com/how-to-cross-validation-with-time-series-data-9802a06272c6/

Model evaluation should reflect the structure of your prediction problem

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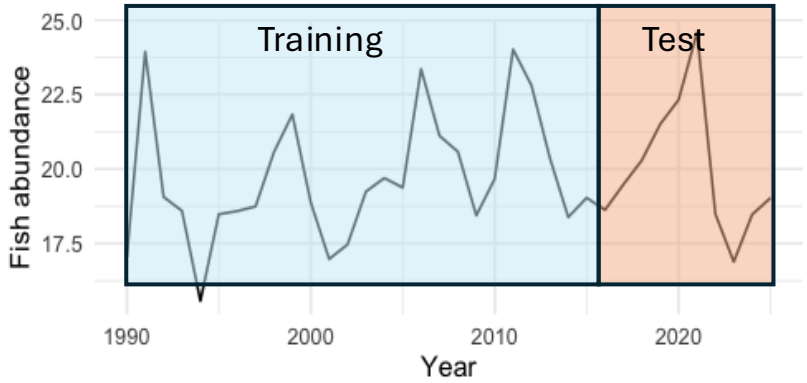
For our fish example, we might be most interested in **predicting the future**



More options for time series:
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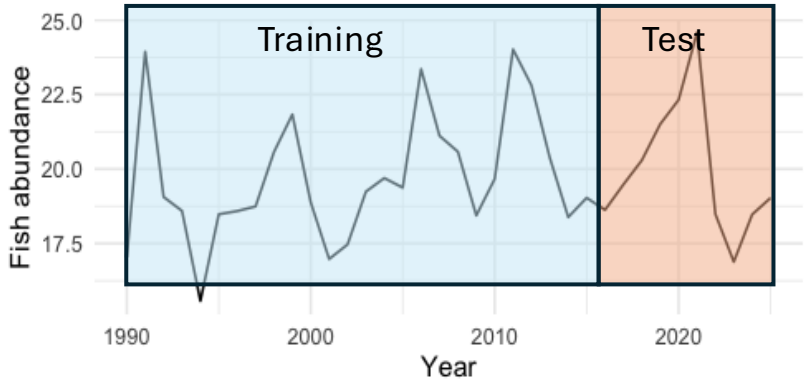

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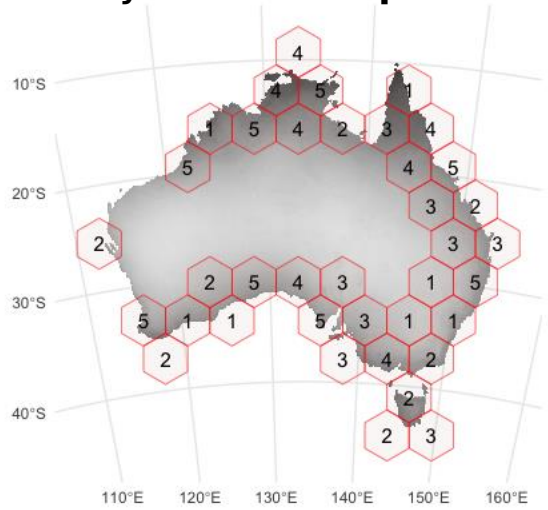
If our prediction problem is spatial, we should probably **consider space** in the model evaluation



blockCV: https://github.com/rvalavi/blockcv

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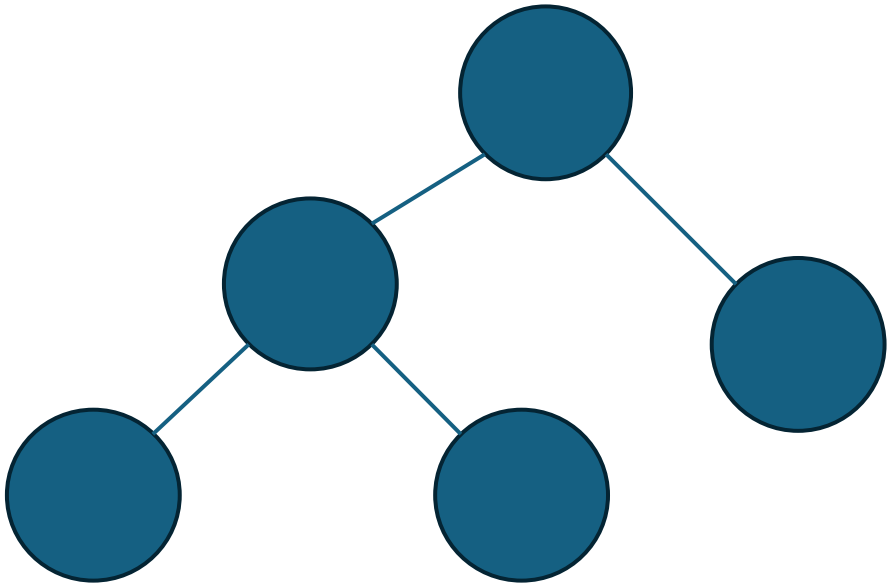
One way that works:

- Build a model using the training set(s)
- Evaluate the model using the test set(s)
- Re-build the model
  - Full dataset
  - Same parameters
  - “Final model”
- Make predictions using the final model

Classification and Regression Trees (CART)

Until a stopping condition is met

- Evaluate best split for each predictor
- Select the optimal split
- Create new nodes



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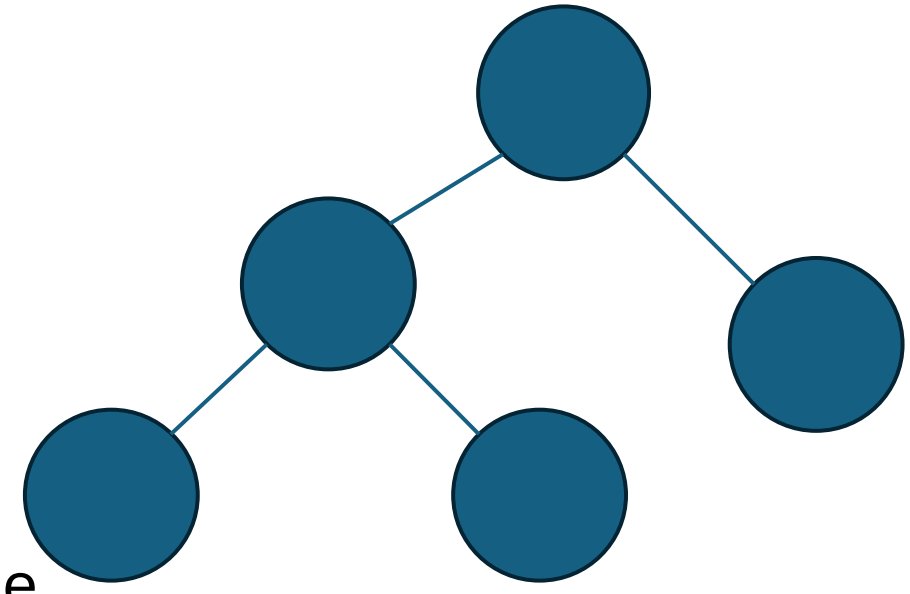


Stopping conditions:

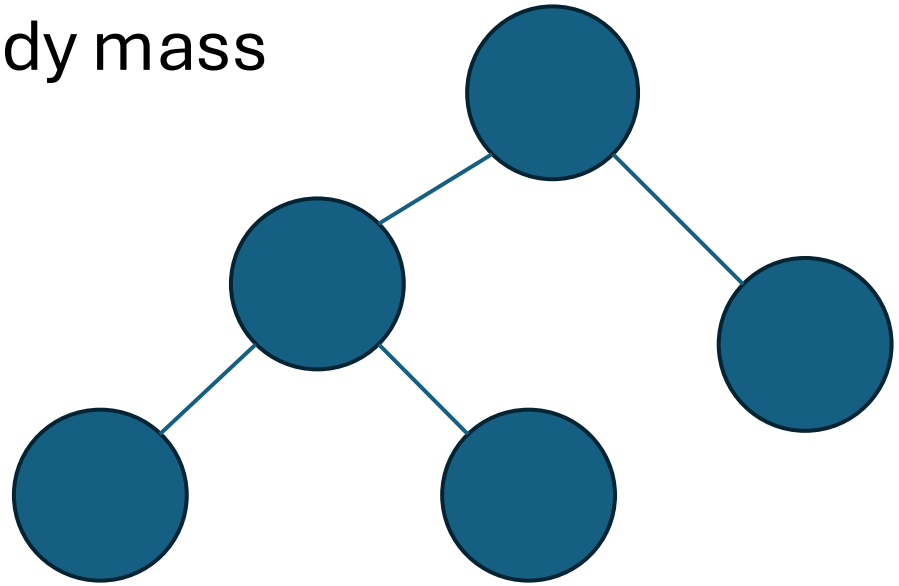
- Maximum tree depth
- Minimum number of samples per node
- No improvement in performance
- Node is all one class / value

Classification and Regression Trees (CART)

Example:

Penguin species ~ flipper length + body mass

Penguin species	Flipper length	Body mass
Adelie	90	80
Gentoo	75	20
Gentoo	77	32
Chinstrap	76	12



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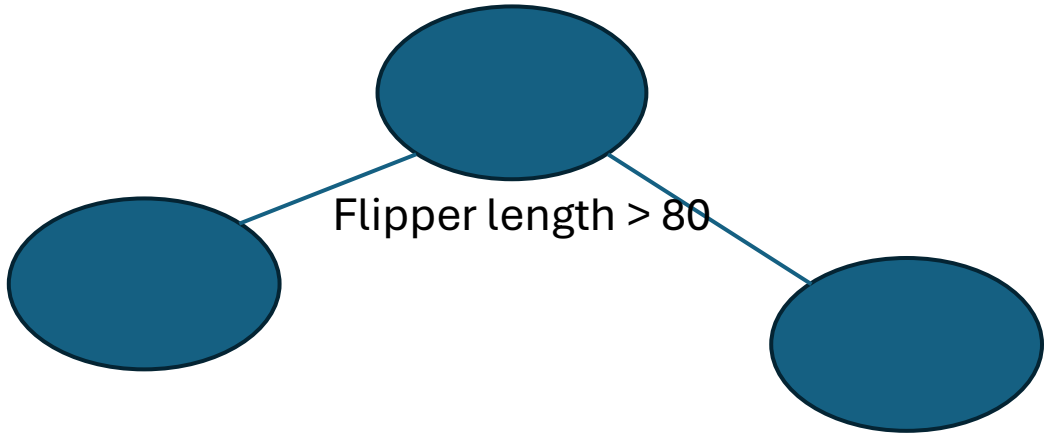

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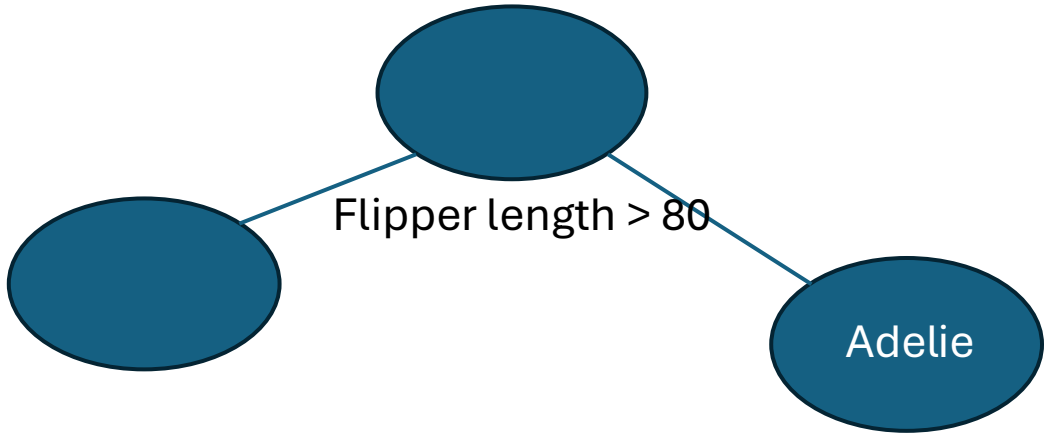

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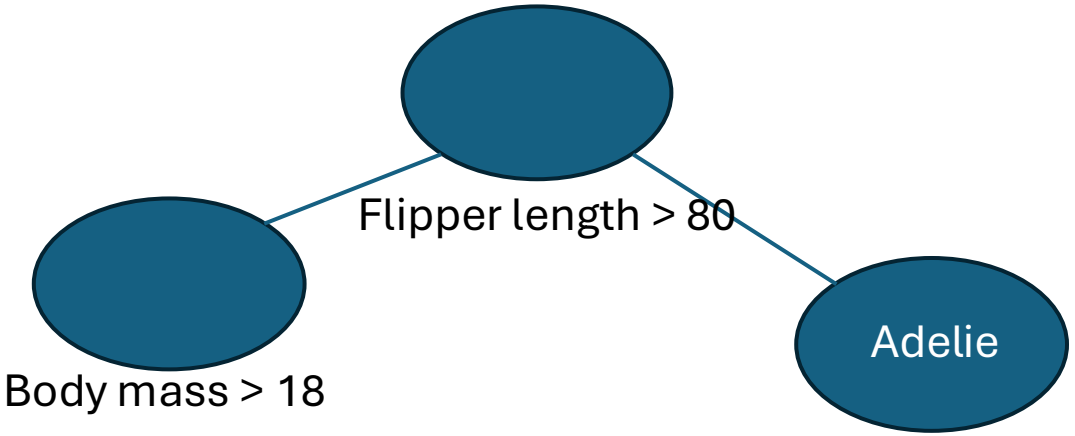

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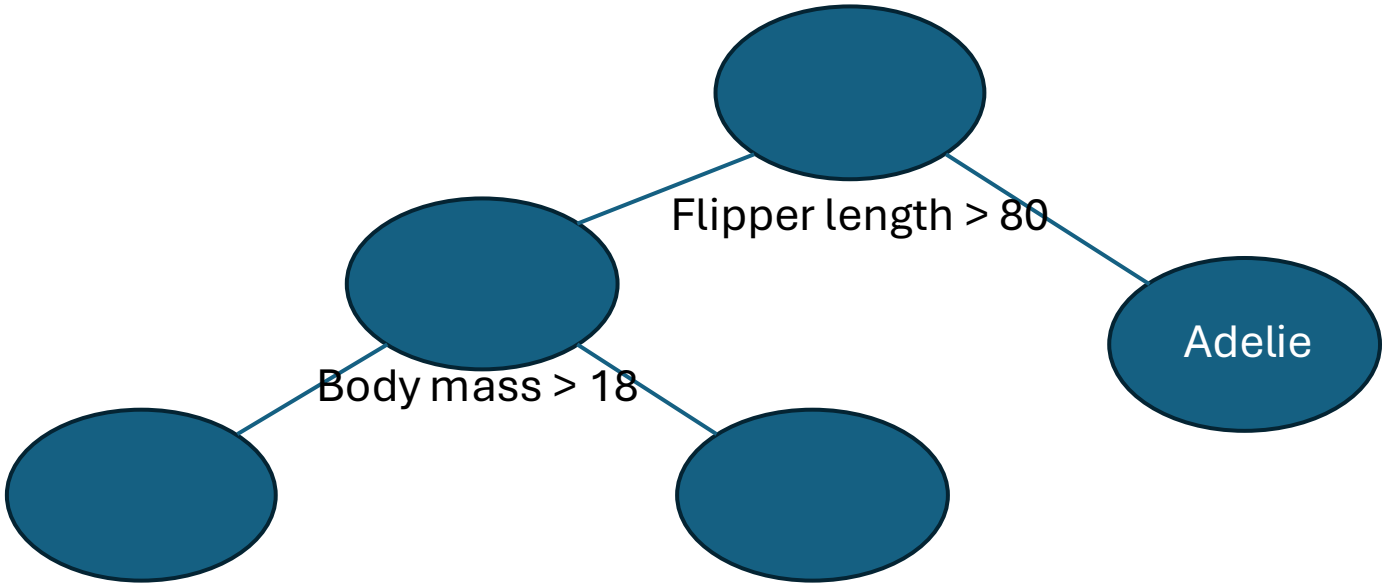

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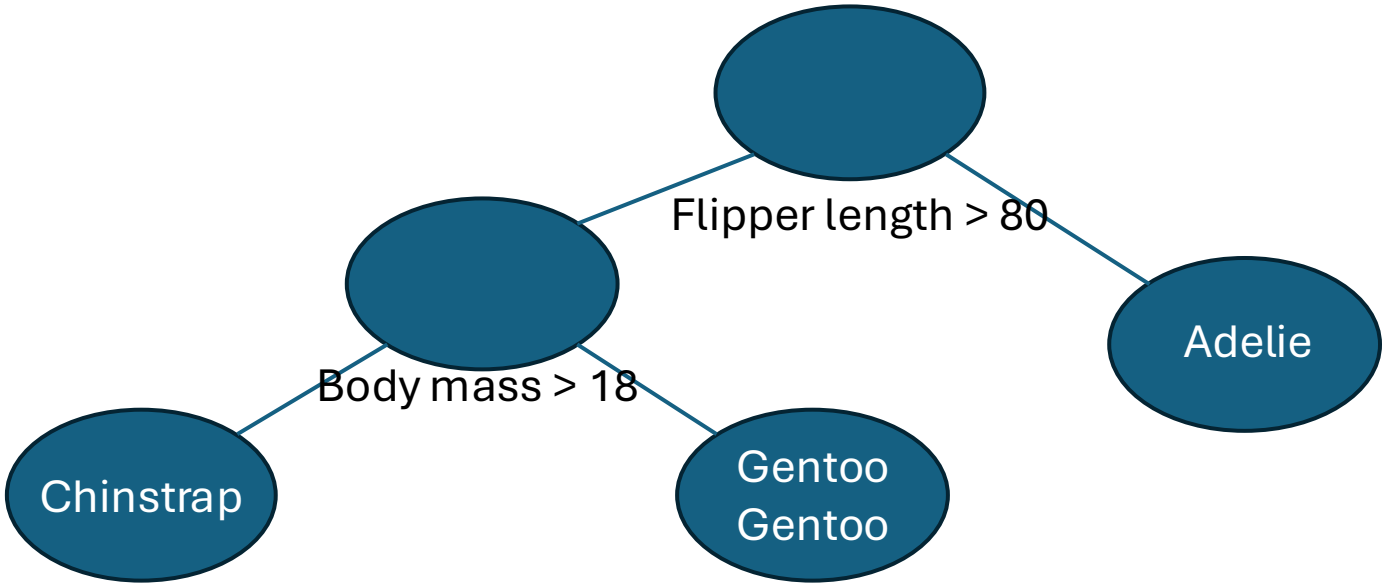
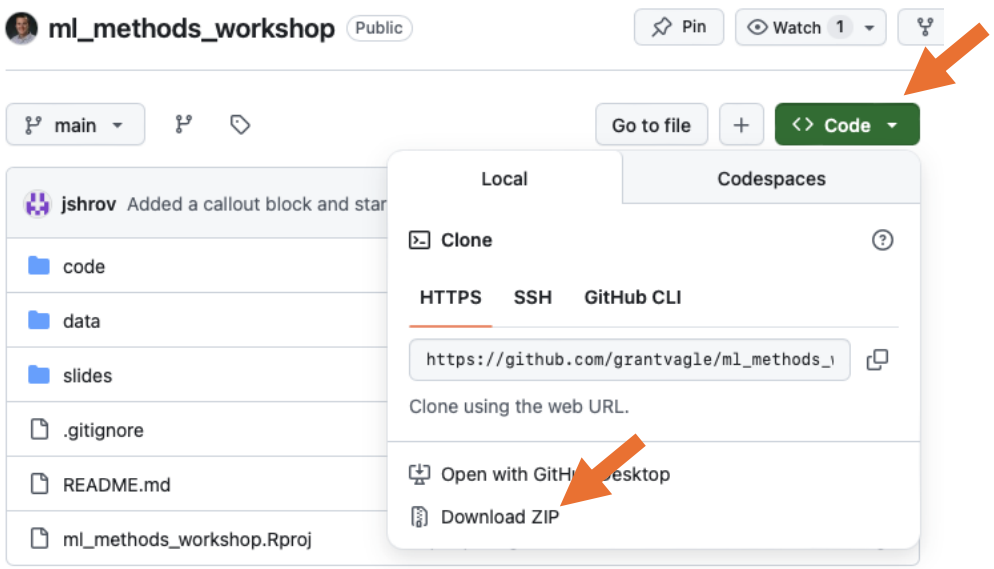
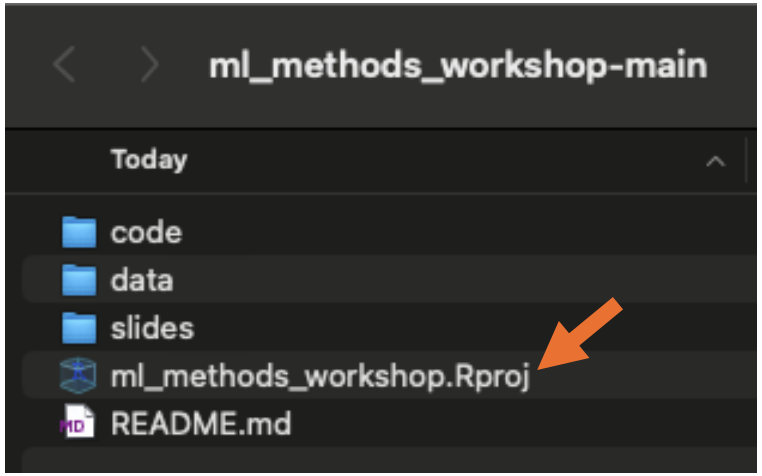


All nodes are “pure” so we stop

Now to a hands-on coding example…

- Go to: https://github.com/grantvagle/ml_methods_workshop
- Click the green “Code” button
- Download the .zip file
- Unzip the folder
- Click to open the .Rproj file





Once you’re set up, work through the **01_cart_examples.qmd** file in the **code** folder